

S&DS 365 / 665  
Intermediate Machine Learning  
**Variational Inference**

October 12

Yale

# Reminders

- Assignment 2 due Thursday
- Midterm week from today, in class
- Material up to and including today's class
- Cheat sheet: One side of 8.5x11 sheet, handwritten
- Quiz 3 scores posted

## Quiz Summary

Section Filter ▾

Student Analysis

Item Analysis

Ⓜ Average Score

**96%**

Ⓢ High Score

**100%**

Ⓣ Low Score

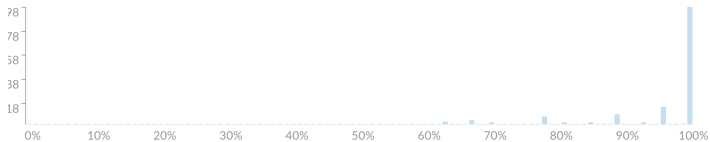
**63%**

Ⓢ Standard Deviation

**0.76**

⌚ Average Time

**14:55**



# Review sessions

- ▶ Thursday Oct 15 (Alice, 4pm)
- ▶ Friday Oct 16 (Ben, 1pm)

# For Today

- Variational inference: The ELBO
- Derivations and examples
- Next time: Variational autoencoders (VAEs)



# Recall: Inverting generative models

Template for generative model:

- 1 Choose  $Z$
- 2 Given  $z$ , generate (sample)  $X$

We often want to invert this:

- 1 Given  $x$
- 2 What is  $Z$  that generated it?



# Inverting models

Bayesian setup:

- 1 Choose  $\theta$
- 2 Given  $\theta$ , generate (sample)  $X$

Posterior inference:

- 1 Given  $x$
- 2 What is  $\theta$  that generated it?



# Inverting models

Often we have two types of latent (hidden) variables, which we'll denote by  $(\theta, z)$ .

The variable  $\theta$  will typically be continuous while  $z$  is discrete.

The goal is to approximate  $p(\theta, z | x)$ .



# Approximate inference

If we have a random vector  $Z \sim p(Z \mid x)$ , we might want to compute the following:

- marginal probabilities  $\mathbb{P}(Z_i = z \mid x)$
- marginal means  $\mathbb{E}(Z_i = z \mid x)$
- most probable assignments  $z^* = \arg \max_z \mathbb{P}(\{Z_i = z_i\} \mid x)$
- maximum marginals  $z_i^* = \arg \max_{z_i} \mathbb{P}(Z_i = z_i \mid x)$
- joint probability  $\mathbb{P}(Z \mid x)$
- joint mean  $\mathbb{E}(Z \mid x)$

Each of these quantities is intractable to calculate exactly, in general.

# Gibbs sampler

Generic algorithm:

- 1 Sample one component holding others fixed
- 2 Iterate

# Gibbs sampler

Generic algorithm:

- 1 Choose index  $j$  randomly
- 2 Sample  $\theta_j$  fixing  $\theta_i, i \neq j$  and  $x$
- 3 Iterate

# Gibbs sampler

Generic algorithm:

- 1 Choose index  $j$  randomly
- 2 Sample  $\theta_j$  from  $\mathbb{P}(\theta_j \mid \theta_i, i \neq j, x)$ .
- 3 Iterate

When run for sufficiently long, this will draw a sample from the posterior  $\mathbb{P}(\theta \mid x)$ .

# Variational methods

- Gibbs sampling is *stochastic* approximation
- Variational methods iteratively make *deterministic* approximations
- Variational and Markov chain approximations originated in physics

# Stochastic approximation–Ising model

## Gibbs sampler

Iterate until converged:

- 1 Choose vertex  $s \in V$  at random
- 2 Sample  $z_s$  holding others fixed

$$\theta_s = \text{sigmoid} \left( \beta_s + \sum_{t \in N(s)} \beta_{st} z_t \right)$$
$$Z_s \mid \theta_s \sim \text{Bernoulli}(\theta_s)$$

You should verify that this is the correct conditional distribution of  $Z_s$  holding the others fixed.

# Deterministic approximation

## Mean field variational algorithm

Iterate until converged:

- 1 Choose vertex  $s \in V$  at random
- 2 Update mean  $\mu_s$  holding others fixed

$$\mu_s = \text{sigmoid} \left( \beta_s + \sum_{t \in N(s)} \beta_{st} \mu_t \right)$$

# Deterministic vs. stochastic approximation

- The  $z_s$  variables are random
- The  $\mu_s$  variables are deterministic
- The Gibbs sampler convergence is in distribution
- The mean field convergence is numerical
- The Gibbs sampler approximates the full distribution
- The mean field algorithm approximates the mean of each node

## Example 2: A finite mixture model

Fix two distributions  $F_0$  and  $F_1$ , with densities  $f_0(x)$  and  $f_1(x)$ , and form the mixture model

$$\begin{aligned}\theta &\sim \text{Beta}(\alpha, \beta) \\ X | \theta &\sim \theta F_1 + (1 - \theta) F_0.\end{aligned}$$

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The likelihood for data  $x_1, \dots, x_n$  is

$$p(x_{1:n}) = \int_0^1 \text{Beta}(\theta | \alpha, \beta) \prod_{i=1}^n (\theta f_1(x_i) + (1 - \theta) f_0(x_i)) d\theta.$$

Our goal is to approximate the posterior  $p(\theta | x_{1:n})$

# Stochastic approximation

## Gibbs sampler

- 1 Sample  $Z_i \mid \theta, x_{1:n}$  for  $i = 1, \dots, n$
- 2 Sample  $\theta \mid z_{1:n}, x_{1:n}$

The first step is carried out by sampling

$$Z_i = \begin{cases} 1 & \text{with probability } \propto \theta f_1(x_i) \\ 0 & \text{with probability } \propto (1 - \theta) f_0(x_i) \end{cases}$$

# Stochastic approximation

## Gibbs sampler

- 1 Sample  $Z_i \mid \theta, x_{1:n}$  for  $i = 1, \dots, n$
- 2 Sample  $\theta \mid z_{1:n}, x_{1:n}$

The second step is carried out by sampling

$$\theta \sim \text{Beta} \left( \sum_{i=1}^n z_i + \alpha, n - \sum_{i=1}^n z_i + \beta \right).$$

Posterior over  $\theta$  is approximated as *mixture* of Beta distributions;  
number of components is  $n + 1$

# Variational inference: Strategy

- We'd like to compute  $p(\theta, z | x)$ , but it's too complicated.
- Strategy: Approximate as  $q(\theta, z)$  that has a “nice” form
- $q$  is a function of “variational parameters” optimized for *each*  $x$ .
- Maximize a lower bound on  $p(x)$ .

# Variational inference: The ELBO

The ELBO is the following lower bound on  $\log p(x)$ :

$$\log p(x) = \int \sum_z q(z, \theta) \log p(x) d\theta$$

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We maximize this over the parameters of  $q$



# Variational inference: The ELBO

The inequality above uses concavity of the logarithm:

$$\log \left( \sum_{\alpha} w_{\alpha} x_{\alpha} \right) \geq \sum_{\alpha} w_{\alpha} \log x_{\alpha}$$

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So, if  $q_{\alpha} \geq 0$  and  $p_{\alpha} \geq 0$  sum (or integrate) to one, then

$$0 = \log \left( \sum_{\alpha} p_{\alpha} \right) = \log \left( \sum_{\alpha} q_{\alpha} \frac{p_{\alpha}}{q_{\alpha}} \right) \geq \sum_{\alpha} q_{\alpha} \log \left( \frac{p_{\alpha}}{q_{\alpha}} \right)$$

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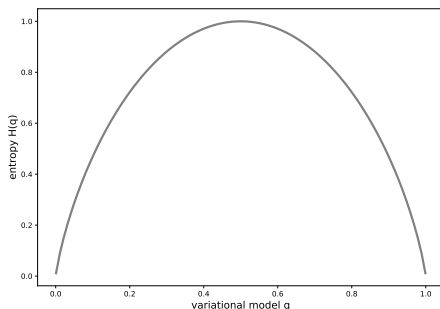
Therefore

$$\sum_{\alpha} q_{\alpha} \log \left( \frac{q_{\alpha}}{p_{\alpha}} \right) \geq 0$$

# Variational inference: The ELBO

The ELBO is  $H(q) + \mathbb{E}_q(\log p)$

The entropy term  $H(q)$  encourages  $q$  to be spread out:



The cross-entropy  $\mathbb{E}_q \log p$  tries to match  $q$  to  $p$

# Variational inference: Strategy

- Choose a simple for the variational approximate posterior  $q_{\gamma}(\theta, z | x)$ .
- Optimize the ELBO over the parameters  $\gamma$

## Example 2: A finite mixture model

Fix two distributions  $F_0$  and  $F_1$ , with densities  $f_0(x)$  and  $f_1(x)$ , and form the mixture model

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$$p(x_{1:n}) = \int_0^1 \text{Beta}(\theta | \alpha, \beta) \prod_{i=1}^n (\theta f_1(x_i) + (1 - \theta) f_0(x_i)) d\theta.$$

Our goal is to approximate the posterior  $p(\theta | x_{1:n})$

# Variational approximation

Our variational approximation is

$$q(z, \theta) = q(\theta \mid \gamma_1, \gamma_2) \prod_{i=1}^n q_i^{z_i} (1 - q_i)^{(1-z_i)}$$

where  $q(\theta \mid \gamma_1, \gamma_2)$  is a  $\text{Beta}(\gamma_1, \gamma_2)$  distribution, and  $0 \leq q_i \leq 1$  are  $n$  free parameters.

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Need to maximize ELBO  $H(q) + \mathbb{E}_q \log p$

Let's sketch part of the calculation

# Variational approximation

First, we have

$$\log p(x, \theta, z) = \log p(\theta \mid \alpha, \beta) + \sum_{i=1}^n \left\{ \log (\theta f_1(x_i))^{z_i} + \log ((1 - \theta) f_0(x_i))^{(1-z_i)} \right\}$$

# Variational approximation

Next, take expectation with respect to  $q$  and use identities such as

$$\mathbb{E}_q \log \theta = \psi(\gamma_1) - \psi(\gamma_1 + \gamma_2)$$

for the digamma function  $\psi(\cdot)$ .

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So, for example,

$$\begin{aligned}\mathbb{E}_q \log (\theta f_1(x_i))^{z_i} &= q_i \mathbb{E}_q \log \theta + q_i \log f_1(x_i) \\ &= q_i (\psi(\gamma_1) - \psi(\gamma_1 + \gamma_2)) + q_i \log f_1(x_i)\end{aligned}$$

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After some calculus and algebra, we end up with the following algorithm (see the notes on the course web page for more detail)

# Variational algorithm for mixture

## Variational inference

Iterate the following steps for variational parameters  $q_{1:n}$  and  $(\gamma_1, \gamma_2)$ :

- 1 Holding  $q_i$  fixed, set  $\gamma = (\gamma_1, \gamma_2)$  to

$$\gamma_1 = \alpha + \sum_{i=1}^n q_i \quad \gamma_2 = \beta + n - \sum_{i=1}^n q_i$$

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$$\gamma_1 = \alpha + \sum_{i=1}^n q_i \quad \gamma_2 = \beta + n - \sum_{i=1}^n q_i$$

- 2 Holding  $\gamma_1$  and  $\gamma_2$  fixed, set  $q_i$  to

$$q_i = \frac{f_1(x_i) \exp \psi(\gamma_1)}{f_1(x_i) \exp \psi(\gamma_1) + f_0(x_i) \exp \psi(\gamma_2)}$$

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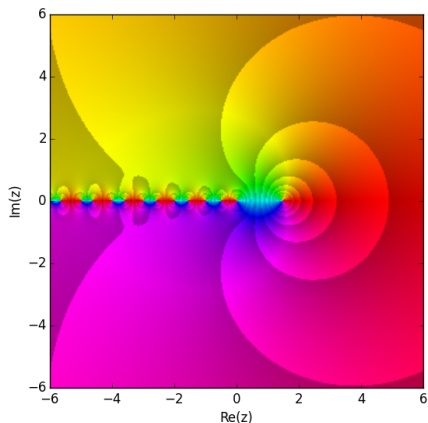
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$$q_i = \frac{f_1(x_i) \exp \psi(\gamma_1)}{f_1(x_i) \exp \psi(\gamma_1) + f_0(x_i) \exp \psi(\gamma_2)}$$

After convergence, approximate posterior distribution over  $\theta$  is

$$\hat{p}(\theta \mid x_{1:n}) = \text{Beta}(\theta \mid \gamma_1, \gamma_2)$$

# Digamma function



$\psi(x)$  is the *digamma function*

[https://en.wikipedia.org/wiki/Digamma\\_function](https://en.wikipedia.org/wiki/Digamma_function)

# Deterministic approximation

- Convergence is numerical, not stochastic
- Posterior is approximated as a *single* Beta
- Very similar algorithm is used for topic models

## Example 3: More general mixtures

$$\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

$$X | \theta \sim \theta_1 F_1 + \dots + \theta_k F_k$$

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$$\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$$

$$X | \theta \sim \theta_1 F_1 + \dots + \theta_k F_k$$

The likelihood for single data point  $x$  is

$$p(x) = \int \text{Dirichlet}(\theta | \alpha_1, \dots, \alpha_k) \left( \sum_{j=1}^k \theta_j f_j(x) \right) d\theta.$$

When distributions  $F_j$  are learned, this is a “topic model.” Variational inference is one of the most useful ways of training topic models

- Notes on variational methods: Please read Sections 1–4
- The other sections are more advanced / specialized material
- Next up: Variational autoencoders
- VAEs not on midterm, but please review basics of variational methods discussed so far, including the ELBO

## **Some questions about variational inference**

# Variational inference

Q: What is the best  $q$  we could use?

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A: The true posterior  $q(z, \theta | x) = p(z, \theta | x)$

# Variational inference

Q: What is the best  $q$  we could use?

A: The true posterior  $q(z, \theta | x) = p(z, \theta | x)$

Why? Because this maximizes the ELBO. Mathematically,

$$\sum_z \int q(z, \theta) \log \left( \frac{q(z, \theta)}{p(z, \theta | x)} \right) d\theta = 0$$

in this case, so the ELBO inequality is an equality.

# Variational inference

Q: How does the ELBO regularize?

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A: The entropy term favors distributions that are “spread out”

# Variational inference

Q: How does the ELBO regularize?

A: The entropy term favors distributions that are “spread out”

Why? For discrete distributions the maximum entropy distribution is uniform. For Gaussian, the entropy is  $\log \sigma^2$  which favors  $\sigma^2$  large.

# Variational inference

Q: Is the ELBO easy to maximize?

# Variational inference

Q: Is the ELBO easy to maximize?

A: No.

# Variational inference

Q: Is the ELBO easy to maximize?

A: No.

Why? In general it is non-convex, and the solution depends on where we start an iterative algorithm. This is unlike the Gibbs sampler, which converges to the right thing if we wait long enough.

# Next class: Variational autoencoders

- Variational autoencoders are generative models that are trained using variational inference
- The “decoder” is a neural net that generates from a latent variable
- The “encoder” approximates the posterior distribution with another neural network trained using variational inference

# Summary

- Gibbs sampling makes stochastic approximations
- Variational methods make deterministic approximations
- General recipe: Maximize ELBO over variational parameters
- Gives a powerful approach to generative modeling